INCH-POUND

L-P-380D
17 February 2026
SUPERSEDING
L-P-380C
August 24, 1973

FEDERAL SPECIFICATION

PLASTIC MOLDING MATERIAL, POLY(METHYL METHACRYLATE) (PMMA)

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers poly(methyl methacrylate) molding compounds which are suitable or compression, injection, and extrusion molding.

1.2 Classification.

1.2.1 Types and Classes. Poly (methyl methacrylate) molding and extrusion compounds covered by this specification should be according to the following types and classes as specified (see 6.2).

Type I	-	General purpose.
Type II	-	Electrical type material.
Class 1	-	Lowest deflection temperature range.
Class 2	-	Middle deflection temperature range.
Class 3	-	Highest deflection temperature range.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2. Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-123	-	Marking for Shipment (Civilian Agencies).
FED-STD-313	-	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities.

(Copies of this document are available online at https://quicksearch.dla.mil/).

FEDERAL SPECIFICATIONS

CCC-C-430	-	Cloth, Sheeting, Cotton, and Polyester and Cotton
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

FEDERAL STANDARDS

FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-313	-	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities.

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM D149	-	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D150	-	Standard Test Methods for AC Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulation
ASTM D257	-	Standard Test Methods for DC Resistance or Conductance of Insulating Materials
ASTM D542	-	Standard Test Method for Index of Refraction of Transparent Organic Plastics
ASTM D570	-	Standard Test Method for Water Absorption of Plastics
ASTM D618	-	Standard Practice for Conditioning Plastics for Testing
ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D648	-	Standard Test Method for Deflection Temperature of Plastics Under Flexural Load in the Edgewise Position
ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1003	-	Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics
ASTM G21	-	Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

(Copies of these documents are available online at https://www.astm.org/).

SAE INTERNATIONAL STANDARDS / Aerospace Material Specifications

SAE-AMS-STD-595	-	Colors Used in Government Procurement.

(Copies of these documents are available from https://www.sae.org).

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2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The methacrylate material shall consist of virgin compression, injection or extrusion molding compounds to which no scrap or reworked material has been added. The methacrylate material shall consist of resins made by polymerization of acrylic monomers containing at least 80 percent methyl methacrylate. Pelletizing shall not be considered reworking.

3.2 Property values. When tested as specified in the applicable procedure of 4.3, type I material shall conform to the property values specified in tables I and II.

TABLE I. Property values, excluding electrical properties.

Property	Property values for types I and II		
	Class 1	Class 2	Class 3
Specific gravity (g/cm^3) 23°C (73.4°F)			
Maximum	1.19	1.19	1.19
Minimum	1.18	1.18	1.18
Index of refraction 1/ (N_D 23°C)			
Maximum	1.50	1.50	1.50
Minimum	1.48	1.48	1.48
Total luminous transmittance 2/ 0.125 inch thickness, percent, minimum	90	90	90
Water absorption, 24 hour immersion, percent, maximum:			
Weight gain plus soluble matter loss	0.5	0.5	0.5
Soluble matter loss	0.1	0.1	0.1
Defection temperature under load 3/ at: 264 pounds per square inch (1.8 MPa) stress			
°C (°F) minimum	67 (153)	76 (169)	85 (185)
°C (°F) maximum	75 (167)	84 (183)	---
0.25 inch thick annealed specimen:			
°C (°F) minimum	69 (156)	80 (176)	87 (189)
°C (°F) maximum	79 (174)	86 (187)	---
Tensile strength			
ksi (MPa) minimum	8.0 (55.2)	8.0 (55.2)	8.0 (55.2)

1/ Applicable only to colorless and to those transparent colors which transmit at the sodium D line. Otherwise, a correction must be made.

2/ Applicable to colorless material only. Correction must be made for colored material.

3/ Either thickness may be used for determining this property.

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TABLE II. Electrical property values (for type II only).

Property	Property value for type II, all classes
Insulation resistance, megohms, minimum	10^7
Dielectric strength, flatwise, step-by-step, volts per mil., minimum	350
Dielectric constant, maximum: at 60 Hz 1/ at 1 kHz	 4.5 4.0
Dissipation factor, maximum: at 60 Hz 1/ at 1 kHz	 0.06 0.05

1/ 1 hertz equals 1 cycle per second.

3.3 Fungus resistance. When specified by the procuring agency (see 6.2), the material shall be fungus resistant as shown by zero percent growth when examined visually.

3.4 Color. Unless otherwise specified by the procuring agency, the material and specimens or plaques molded therefrom shall be colorless as determined by visual inspection. When required by the procuring agency (see 6.2) a specific method for determining colorless material shall be specified by the agency. When color is specified (see 6.2) it shall be determined visually by comparison of molded plaques with the appropriate color number shown in SAE-AMS-STD-595.

3.5 Form and uniformity. The form shall be as specified by the procuring agency. (see 6.3), Each lot of molding and extruding compound shall be uniform in form.

3.6 Residual monomer. The amount of residual poly(methyl methacrylate) monomer in plastic molding material, poly(methyl methacrylate), shall not exceed 0.5 percent by weight.

3.7 Workmanship. Each lot of molding and extruding compound shall be clean and free of foreign particles and other contamination.

3.8 Markings. Containers of Methacrylate molding and extrusion compounds covered by this specification shall include labels that identify the specification number, type, and class so that traceability of the lot can be verified and include a SDS (see 3.9 and 6.4).

3.9 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief:

(http://www.osha.gov/dsg/hazcom/osha-brief.html).

and, as specified in Appendix D of 29 CFR 1910.1200 see:

(https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200AppD).

A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the qualifying activity (see 6.4.3). The SDS shall be included with each shipment of the material

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covered by this specification and submitted to pertinent Government agencies as stated in FED-STD-313.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use their own or any other laboratory facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-1916, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type and class submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for ensuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with 3.1 and 3.5 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the list of defects, inspection levels, and acceptable quality levels (AQL) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-1916, shall be expressed in units of 300 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination, specified in table III, shall be approximately one pound.

TABLE III. Examination of the material for defects in appearance and workmanship.

Examine	Defects
Appearance and workmanship	Not colorless, unless otherwise specified. Color not as specified when color required. Form or color not uniform. Form not as specified. Not clean, presence of foreign material or other contamination.

4.2.2.1.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether the packing and marking comply with the requirements in section 5. Defects

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shall be scored in accordance with table IV. The sample unit for this inspection shall be one shipping container fully prepared for shipment. When specified by the procuring agency, the shipping container shall be selected just prior to the closing operation. Sampling and inspection shall be in accordance with 4.2.2.1.3.

TABLE IV. Examination of preparation for delivery.

Examine	Defects
Packing and packaging (see 5.1)	Not in accordance with contract or purchase order. Any nonconforming component, component missing, damaged, or otherwise defective affecting serviceability. Liner in fiber drums (as applicable) not properly sealed. Closure improper.
Quantity of Materials	Other than specified.
Markings	Markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence, method of application, or not in compliance with the contract requirements.

4.2.2.1.3 Inspection levels and AQL's for examinations. The inspection levels for determining the sample size and the AQL accordance with MIL-STD-1916 and ANSI/ASQ Z1.4 expressed as defects per 100 units are to be finalized in the quality assurance section of the contract or purchase order. Legacy inspection and acceptable quality levels from the previous revision are listed for reference in table V:

TABLE V. Inspection levels. 1/

Examination Paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S2	2.5

1/ Historic levels from L-P-380C listed for reference.

4.2.3 Testing. The material shall be tested for the characteristics listed in tables I and II as applicable, in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 300 pounds. The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1 with an acceptance number of 0 defects. The results for each test shall be the averaged results of the specimens.

4.3 Test methods.

4.3.1 Specimen preparation and conditioning. Specimens shall be prepared by injection molding under conditions specified by the manufacturer.

4.3.1.1 Specimen annealing. Specimens to be used for deflection temperature under load, and tensile strength tests shall be conditioned at the temperatures shown in table VI in a forced air

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circulation oven. Specimens 0.125 inch thick shall be conditioned 2 hours, and specimens 0.25 inch thick shall be conditioned 4 hours. Upon removal from the oven, the specimens shall be placed in an airtight container and protected with a blanket of insulation to allow slow cooling to room temperature. The specimens shall remain in the container until ready to test. (Annealing is optional for specimens to be used for other tests.)

TABLE VI. Annealing temperatures.

Class	Annealing Temperatures	
	(°C)	(°F)
1	60 ± 3	140 ± 5.4
2	71 ± 3	160 ± 5.4
3	82 ± 3	180 ± 5.4

4.3.1.2 Specimen conditioning. Specimens of all tests other than those specified in 4.3.1.1, shall be conditioned in accordance with procedure A of ASTM D618 and the minimum conditioning time shall be 24 hours.

4.3.2 Testing conditions. Unless otherwise specified by the applicable test method, all tests shall be performed at 23 ± 2°C (73.4 ± 3.6°F,) and 50 ± 5 percent relative humidity.

4.3.3 Specific gravity. One specimen shall be tested in accordance with method A-1 or A-2 of ASTM D792.

4.3.4 Index of refraction. One specimen shall be tested in accordance with ASTM D542.

4.3.5 Total luminous transmittance. One specimen, 0.125 ± 0.01 inch thick, shall be tested in accordance with procedure A of ASTM D1003.

4.3.6 Water absorption. One specimen shall be tested in accordance with ASTM D570.

4.3.7 Deflection temperature under load. Three specimens, each either 0.125 ± 0.01 inch thick, or 0.25 ± 0.01 inch thick, shall be tested in accordance with ASTM D648.

4.3.8 Tensile strength. Five specimens, each 0.125 ± 0.01 inch thick shall be tested in accordance with ASTM D638, using testing speed B. Dimensions of each specimen shall conform to type I in ASTM D638.

4.3.9 Insulation resistance (for type II material only). Three specimens shall be tested in accordance with ASTM D257 using the electrodes specified in this test method.

4.3.10 Dielectric strength (for type II material only). Three specimens shall be tested in accordance with ASTM D149, using electrodes as specified in this test method, and the short time test. Tests shall be made under oil which conforms to the parameters specified in ASTM D149.

4.3.11 Dielectric constant and dissipation factor (for type II material only). Three specimens shall be tested in accordance with ASTM D150, using any electrode system specified in this test method.

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4.3.12 Fungus resistance (applicable only when specified by the procuring agency). Three specimens shall be prepared and tested in accordance with ASTM G21. Specimens shall be 2.0 by 2.0 inch pieces as specified in ASTM G21.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Packaging and container notes. Due to the substantial passage of time since the last revision of this specification, additional packaging and container information pertaining to container size and weight that was used in the past is included that may be helpful and is included in 6.5. This legacy content will be deleted in the next revision or amendment to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Methacrylate plastics may be used in applications requiring excellent clarity and transparency, resistance to sunlight and outdoor weathering, and an optimum combination of flexibility with shatter resistance and rigidity. Examples are edge lighted cockpit and interior control panels. The classes having higher heat distortion temperatures may be used for applications requiring greater heat resistance. A chief disadvantage of these plastics is a lack of abrasion resistance and a tendency to craze under stress, particularly when simultaneously exposed to active solvents or wetting agents. The materials are not for shipboard use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Types and classes required (see 1.2.1).
c. Fungus resistance when specified (see 3.3), including type and class of material.
d. When required, specific method for determining colorless material (see 3.4).
e. Color, if required (see 3.4).
f, Form required (see 3.5).
g. Selection of applicable level of packing (see 5.1).
h. If other container is required (see 5.1, 5.3, and 6.5).

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i. Preparation of SDS in accordance with FED-STD-313 for the adhesive and inclusion of SDS with shipment of material (see 3.9 and 6.4).

6.3 Form. Injection molding and extrusion material is supplied, usually in the form of 0.125 inch pellets. For compression molding, material is usually supplied as minute spherical granules.

6.4 SDS. Contracting officers should identify those activities requiring copies of a completed SDS, when implemented, prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Packaging and container considerations. Due to the substantial passage of time since the last revision of this specification, additional packaging and container information is included as a courtesy in this section.

a. Fiber drums conforming in quantities of 200 pounds maximum.

b. 55 gallon steel drums in quantities of 400 pounds maximum.

Insofar as practical, drums should be of uniform shape and size, with minimum cube tare consistent with the protection required. Drums should contain identical quantities and should be closed in accordance with applicable container specifications. To avoid potential fungal nucleation and growth, fiber drums should be furnished with a 0.004 inch thick and heat sealed polyethylene liner.

6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ABC-NAVY-17C. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Subject term (key word) listing.

Base Material	Extrusion	Pellets
Compression Molding	Injection Molding	Polymer

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Air Force - 20

Preparing activity:
ARMY - MR

(Project 9330-2025-004)

Review activities:
Army - EA, MD, MI
Navy - AS, EC, OS, SH

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